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CALCULUS PROBLEM SHEET

1

- (a) Show that if the columns, $\mathbf{m}_j$, of an $n \times n$ matrix, $\mathbf{M}$, are not independent then $\mathbf{M}$ does not have a unique inverse. For the columns to not be independent implies there exists some set of coefficients c_j , not all zero, such that

$$\sum_{j=1}^n c_j \mathbf{m}_j = \mathbf{0}.$$

We note that the i^{th} element of $\mathbf{m}_j = M_{ij}$ so

$$\forall_{i=1}^n : \sum_{j=1}^n c_j M_{ij} = 0$$

[3 marks]

3

- (b) Show that for three $n \times n$ matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ the inverse of $\mathbf{ABC}$ is $\mathbf{C}^{-1}\mathbf{B}^{-1}\mathbf{A}^{-1}$. You can do this by multiplying the product of matrices by its inverse and using associativity of matrix multiplication [3 marks]

3

- (c) For non-square matrices, **A** and **B** we can have $\mathbf{AB} = \mathbf{I}$ while $\mathbf{BA} \neq \mathbf{I}$. Show this is the case for the matrices

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \end{pmatrix} \qquad \mathbf{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

[3 marks]

3

- (d) For finite-dimensional square matrices if $\mathbf{AB} = \mathbf{I}$ then $\mathbf{BA} = \mathbf{I}$. Thus $\mathbf{B} = \mathbf{A}^{-1}$. This is not entirely trivial to prove. We break the proof down. Show that if $\mathbf{AB} = \mathbf{I}$ then **B** is injective (one-to-one). This means that for two different vectors x and y then $\mathbf{B}x \neq \mathbf{B}y$. Equivalently $\mathbf{B}(x - y) \neq \mathbf{0}$, unless $x = y$, but as x and y are arbitrary the only vector for which $\mathbf{B}x = \mathbf{0}$ is the vector $x = \mathbf{0}$. Prove that if $\mathbf{AB} = \mathbf{I}$ and $\mathbf{B}x = \mathbf{0}$ then $x = \mathbf{0}$. [2 marks]

2

- (e) Next we want to show that in a finite dimensional space a square matrix that is injective is also surjective (onto). That is, if $\mathbf{B}$ is injective then for any vector $\mathbf{y}$ there exists a vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{B}\mathbf{x}$. The reason a finite dimensional square matrix that is injective is also surjective is not so obvious. To prove this we consider a set of n **basis** vectors that span the entire space $\mathbb{R}^n$. We call these $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$. Because these are independent the only set of numbers c_i such that $\sum_i c_i \mathbf{v}_i = \mathbf{0}$ is when $c_1 = c_2 = \dots = c_n = 0$. Show that the set of vectors $\mathbf{v}'_i = \mathbf{B}\mathbf{v}_i$ form a set of basis vectors (i.e. they are linearly independent). This is easily proved by contradiction. [5 marks]

5

- (f) Use the fact that the vectors $\mathbf{v}'_i$ are linearly independent to show that, for any vector $\mathbf{y}$, we can find a vector $\mathbf{x}$ such that $\mathbf{y} = \mathbf{B}\mathbf{x}$ [2 marks]

2

(g) Using the fact that $\mathbf{AB} = \mathbf{I}$ and $\mathbf{B}$ is surjective show $\mathbf{BA} = \mathbf{I}$ [2 marks]

2

End of question 1

(a) $\frac{\quad}{3}$	(b) $\frac{\quad}{3}$	(c) $\frac{\quad}{3}$	(d) $\frac{\quad}{2}$	(e) $\frac{\quad}{5}$	(f) $\frac{\quad}{2}$	(g) $\frac{\quad}{2}$	Total $\frac{\quad}{20}$
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2 Throughout this question you may use two standard properties of the trace, namely that a matrix and its transpose have the same trace, and that the trace is unchanged by cyclically permuting a product

$$\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}^T)$$

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA}).$$

(a) For a vector $\mathbf{a} \in \mathbb{R}^m$, a vector $\mathbf{b} \in \mathbb{R}^n$ and an $m \times n$ matrix $\mathbf{U}$, show that

$$\mathbf{a}^T \mathbf{U} \mathbf{b} = \text{tr}(\mathbf{U}^T \mathbf{a} \mathbf{b}^T).$$

The quickest route is to write both sides out in components. Note that $\mathbf{a} \mathbf{b}^T$ is an $m \times n$ matrix while $\mathbf{a}^T \mathbf{U} \mathbf{b}$ is a scalar. [5 marks]

(b) Verify the identity explicitly for

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\mathbf{b} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$\mathbf{U} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

by computing both $\mathbf{a}^T \mathbf{U} \mathbf{b}$ and $\text{tr}(\mathbf{U}^T \mathbf{a} \mathbf{b}^T)$.

[4 marks]

$\frac{4}{1}$

- (c) For a square matrix $\mathbf{U}$ and a sufficiently small scalar ϵ , the inverse of $\mathbf{I} + \epsilon\mathbf{U}$ is given by the geometric (Neumann) series

$$(\mathbf{I} + \epsilon \mathbf{U})^{-1} = \mathbf{I} - \epsilon \mathbf{U} + \epsilon^2 \mathbf{U}^2 - \epsilon^3 \mathbf{U}^3 + \dots.$$

Show this by multiplying $\mathbf{I} + \epsilon \mathbf{U}$ by the truncated series

$$\mathbf{s}_n = \sum_{k=0}^n (-\epsilon)^k \mathbf{u}^k$$

and showing that the result differs from $\mathbf{I}$ only at order ϵ^{n+1} . Hence write down $(\mathbf{I} + \epsilon \mathbf{U})^{-1}$ to first order in ϵ . [5 marks]

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- (d) Use this to find the derivative of $g(\mathbf{X}) = \mathbf{a}^\top \mathbf{X}^{-1} \mathbf{b}$ with respect to the matrix $\mathbf{X}$. Replace $\mathbf{X}$ by $\mathbf{X} + \epsilon \mathbf{U}$, expand to first order in ϵ , and use the definition of the derivative with respect to a matrix

$$\lim_{\epsilon \rightarrow 0} \frac{g(\mathbf{X} + \epsilon \mathbf{U}) - g(\mathbf{X})}{\epsilon} = \text{tr} \left(\mathbf{U}^\top \frac{\partial g(\mathbf{X})}{\partial \mathbf{X}} \right)$$

which holds for every $\mathbf{U}$. Part (a) is what lets you read the derivative off.

[6 marks]

6

End of question 2

(a) $\frac{1}{5}$ (b) $\frac{1}{4}$ (c) $\frac{1}{5}$ (d) $\frac{1}{6}$ Total $\frac{1}{20}$

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3 We want to compute the derivative of $\log(\det(\mathbf{X}))$ with respect to the matrix $\mathbf{X}$. Our strategy will be to use the fact that

$$\lim_{\epsilon \rightarrow 0} \frac{\log(\det(\mathbf{X} + \epsilon \mathbf{U})) - \log(\det(\mathbf{X}))}{\epsilon} = \text{tr} \mathbf{U}^T \frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}.$$

We therefore expand $\log(\det(\mathbf{X} + \epsilon \mathbf{U}))$ to first order in ϵ , subtract $\log(\det(\mathbf{X}))$, divide by ϵ , rearrange and read off $\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}$. However, rearranging in this case is quite complicated.

- (a) Using $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ and $\log(ab) = \log(a) + \log(b)$ expand $\log(\det(\mathbf{X} + \epsilon \mathbf{U}))$ as a sum of $\log(\det(\mathbf{X}))$ and another logarithm. [5 marks]

5

- (b) The hard part is to expand $\det(\mathbf{I} + \epsilon \mathbf{M})$. Write out $\det(\mathbf{I} + \epsilon \mathbf{M})$ explicitly for a 2 dimensional matrix and expand to first order in ϵ [2 marks]

2

(c) For an $n \times n$ matrix $\mathbf{M}$ compute the determinant of

$$\det(\mathbf{I} + \epsilon \mathbf{M}) = \det \begin{pmatrix} 1 + \epsilon M_{11} & \epsilon M_{12} & \dots & \epsilon M_{1n} \\ \epsilon M_{21} & 1 + \epsilon M_{22} & \dots & \epsilon M_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon M_{n,1} & \epsilon M_{n2} & \dots & 1 + \epsilon M_{nn} \end{pmatrix}$$

up to first order in ϵ . Recall we can compute the determinant of a matrix $\mathbf{X}$ as

$$\det(\mathbf{X}) = \sum_{j=1}^n (-1)^{1+j} X_{1j} \det(\hat{\mathbf{X}}_{1j})$$

where $\hat{\mathbf{X}}_{1j}$ is the matrix obtained by removing row 1 and column j from the matrix $\mathbf{X}$. [5 marks]

5

(d) For a more elegant proof we can use the fact that

$$\det(\mathbf{M}) = \prod_{i=1}^n \lambda_i \quad \text{tr} \mathbf{M} = \sum_{i=1}^n \lambda_i$$

where λ_i are the eigenvalues of the matrix $\mathbf{M}$. We will prove these results later when we look at eigen-systems. Note that if $\mathbf{v}_i$ is an eigenvector of the matrix $\mathbf{M}$, with eigenvalue λ_i then it is an eigenvector of $\mathbf{I} + c\mathbf{M}$ with eigenvalue $1 + c\lambda_i$ since

$$(\mathbf{I} + c\mathbf{M})\mathbf{v}_i = \mathbf{v}_i + c\lambda_i\mathbf{v}_i = (1 + c\lambda_i)\mathbf{v}_i.$$

Using these results show that $\det(\mathbf{I} + \epsilon\mathbf{M}) = 1 + \epsilon\text{tr} \mathbf{M} + O(\epsilon^2)$. [3 marks]

3

(e) Using the results

$$\begin{aligned} \log(\det(\mathbf{X} + \epsilon\mathbf{U})) - \log(\det(\mathbf{X})) &= \log(\det(\mathbf{I} + \epsilon\mathbf{X}^{-1}\mathbf{U})) \\ \det(\mathbf{I} + \epsilon\mathbf{M}) &= 1 + \epsilon\text{tr} \mathbf{M} + O(\epsilon^2) \end{aligned}$$

compute $\frac{\partial \log(\det(\mathbf{X}))}{\partial \mathbf{X}}$. You may use the fact that $\log(1 + x) = x + O(x^2)$ and $\text{tr} \mathbf{A} = \text{tr} \mathbf{A}^T$. [5 marks]

5

End of question 3

(a) $\frac{1}{5}$ (b) $\frac{1}{2}$ (c) $\frac{1}{5}$ (d) $\frac{1}{3}$ (e) $\frac{1}{5}$ Total $\frac{1}{20}$

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4 In this question I want you to use a numerical package to do some simple algebra. You can use MATLAB, OCTAVE, numpy in Python or any other language of your choosing. The answers are only going to be accurate up to the numerical precision of your machine (typically around 10^{-16}). For python the following may be useful

<code>import numpy as np</code>	import the numpy module
<code>A = np.random.randn(n,m)</code>	generate a matrix of normally distributed random deviates with n rows and m columns
<code>A.transpose()</code>	return the transpose of A (does not change A)
<code>A@B</code>	matrix multiplication AB . Beware that <code>A*B</code> performs elementwise multiplication which is different to normal matrix multiplication
<code>A.trace()</code>	returns the trace of a matrix
<code>np.linalg.det(A)</code>	return the determinant of a matrix
<code>np.round(A, 10)</code>	rounds the elements in a matrix A to 10 decimal places. Makes expressions like $(A@B)@C - A@(B@C)$ easier to read
<code>np.linalg.inv(A)</code>	inverse of matrix A

Generate three 5×5 random matrices **A**, **B** and **C** and two random vectors of length 5, **a** and **b**.

- (a) Compute **D** = **AB** and **E** = **BC**. Which quantities are the same (a) **ABC**, (b) **DE**, (c) **DC**, (d) **AE**, (e) **CD** [2 marks]

$\frac{2}{2}$

- (b) Compute **AB** and **BA**. Are these the same? [2 marks]

$\frac{2}{2}$

- (c) Compute (a) $a^T A b$, (b) $a^T A^T b$ (c) $b^T A^T a$ and (d) $b^T A a$. Which quantities are the same? [2 marks]

$\frac{2}{2}$

- (d) Compute (a) $\text{tr} A B$, (b) $(\text{tr} A)(\text{tr} B)$ (c) $\text{tr} B A$ and (d) $\text{tr} B^T A^T$. Which quantities are the same? [2 marks]

$\frac{2}{2}$

- (e) Compute (a) $\det(A B)$, (b) $(\det A)(\det B)$ (c) $\det(B A)$ and (d) $\det(B^T A^T)$. Which quantities are the same? [2 marks]

$\frac{2}{2}$

(f) Although we can define the gradient with respect to a matrix using

$$\lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{A} + \epsilon \mathbf{U}) - f(\mathbf{A})}{\epsilon} = \text{tr} \mathbf{U}^T \frac{\partial f(\mathbf{A})}{\partial \mathbf{A}}$$

numerically the convergence is slow. This is problematic as $f(\mathbf{A} + \epsilon \mathbf{U})$ and $f(\mathbf{A})$ are very close in value so when we take their difference we lose a lot of precision. A ratio that converges much more quickly is

$$\lim_{\epsilon \rightarrow 0} \frac{f(\mathbf{A} + \epsilon \mathbf{U}) - f(\mathbf{A} - \epsilon \mathbf{U})}{2\epsilon} = \text{tr} \mathbf{U}^T \frac{\partial f(\mathbf{A})}{\partial \mathbf{A}}.$$

If we want to obtain a more accurate numerical estimate of a derivative we would compute the ratio for different values of ϵ and then extrapolate this series to when $\epsilon \rightarrow 0$. However, for this exercise this is unnecessary. Let $f(\mathbf{A}) = \log(\det(\mathbf{A}))$, set $\mathbf{U} = \mathbf{B}$ and compute

$$\text{tr} \mathbf{U}^T \frac{\partial f(\mathbf{A})}{\partial \mathbf{A}}$$

numerically. Compare this to $\text{tr} \mathbf{U}^T (\mathbf{A}^{-1})^T$

[10 marks]

10

End of question 4

(a) $\frac{2}{2}$	(b) $\frac{2}{2}$	(c) $\frac{2}{2}$	(d) $\frac{2}{2}$	(e) $\frac{2}{2}$	(f) $\frac{10}{10}$	Total $\frac{20}{20}$
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